Syllabus for Part 2

TECH6: Data Science and Econometrics

Richard Foltyn

NHH Norwegian School of Economics

October 13, 2026

See the GitHub repository for notebooks and data:
https://github.com/richardfoltyn/TECH6-H26

Contact and course materials

Contact

- Email: richard.foltyn@nhh.no
- Office: D231, 2nd floor in the new building

Course materials

- GitHub repository (notebooks and data): https://github.com/richardfoltyn/TECH6-H26
- PDFs also posted on Canvas

Teaching

Lectures

- Tuesdays, 10:15–12:00, Aud O
- Hands-on introduction to concepts and methods in Python

Workshops

- Thursdays, **8:30**–10:00, Aud O
- Apply the lecture material to exercises on your own laptop

Part 2: Course outline

Week 42: Introduction to scikit-learn

- Linear regression; training and test samples
- Pipelines, polynomial features, and categorical data

Week 43: Cross-validation and model selection

- Bias-variance trade-off (underfitting vs. overfitting)
- Cross-validation and hyperparameter tuning
- Principal component regression

Part 2: Course outline (continued)

Week 44: Regression models

- Regularization: Ridge, Lasso, and Elastic Net
- Feature scaling and tuning regularization parameters

Week 45: Classification models

- Logistic regression, decision trees, and random forests
- Cross-validation with grid search

Mandatory assignment

- Handout date: Thursday, October 29 at 9:00
- Due: Thursday, November 5 at 12:00
- Submit on Canvas
- Grade: Pass/fail
- The assignment is to be completed individually

Literature

- Trevor Hastie, Robert Tibshirani, and Jerome Friedman: *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd edition.
  https://hastie.su.domains/ElemStatLearn/

- scikit-learn User Guide:
  https://scikit-learn.org/stable/user_guide.html